Křivočaré souřadnice

M … prostor

$\mathcal{F}M$ … funkce na M $\qquad \mathcal{F}M \ni f : M \to \mathbb{R}$

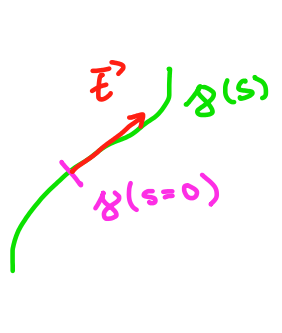
$\mathcal{T}M$ … vektorová pole na M $\qquad \mathcal{T}M \ni \vec{t} : \vec{t}|_m \in \mathbb{V}$

$\mathcal{T}^*M$ … kovektorová pole na M $\qquad \mathcal{T}^*M \ni \underset{\sim}{\alpha} : \underset{\sim}{\alpha}|_m \in \mathbb{V}^*$

$\mathcal{T}^k_l M$ … tenzorová pole na M $\qquad \mathcal{T}^k_l M \ni T : M \to T^k_l$

tenzor typu (k,l)

M … souřadnice x^i $\quad i \in \{1, \dots, \dim M\}$



$s \in \mathbb{R}$

$$\vec{t} = \frac{d\gamma}{ds}\Big|_{s=0}$$

$$f \in \mathcal{F}M \qquad D_{\vec{t}} f = \frac{df(\gamma(s))}{ds} = (\vec{t} \cdot \nabla) f$$

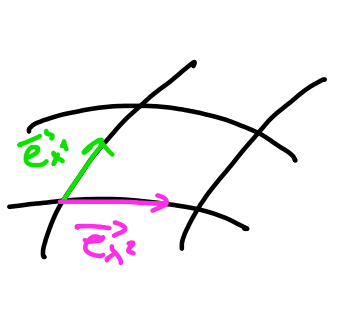
směrová derivace

gradient derivace f podle směru

nezávisí na γ, ale pouze na tečně

$$\Rightarrow \quad D_{\vec{t}} \longleftrightarrow \vec{t}$$

je obousměrná!



$$D_{\vec{e}_{x^1}} f = \frac{df(x^1+s, x^2, \dots)}{ds} = \frac{\partial f}{\partial x^1}$$

$$\Rightarrow \quad D_{\vec{e}_{x^j}} = \frac{\partial}{\partial x^j}$$

$$\Rightarrow \quad \left\{\frac{\partial}{\partial x^j}\right\}_{i=1}^{\dim M} \quad \dots \text{ báze } \mathcal{T}M$$

$\underset{\sim}{\alpha} \in \mathcal{T}^*M$, $\vec{t} \in \mathcal{T}M$ $\qquad \underset{\sim}{\alpha}(\vec{t}) \in \mathbb{R}$

$df \in \mathcal{T}M \qquad df(\vec{t}) = \frac{\partial f}{\partial x^i} t^i$

$x^i : M \to \mathbb{R} \quad \Rightarrow \quad dx^i \quad \Rightarrow \quad \{dx^i\}_{i=1}^{\dim M}$

$$dx^i\left(\frac{\partial}{\partial x^j}\right) = \frac{\partial x^i}{\partial x^k}\left(\frac{\partial}{\partial x^j}\right)^k = \delta^i_j$$

$\Rightarrow$ báze jsou duální

Transformace

$x^i \to x'^j$

$$\frac{\partial}{\partial x'^j} = \frac{\partial x^i}{\partial x'^j}\frac{\partial}{\partial x^i} \qquad dx'^j = \frac{\partial x'^j}{\partial x^i} dx^i$$

$$T(\vec{t}, \underset{\sim}{\alpha}) = T\left(t^i \frac{\partial}{\partial x^i}, \alpha_j dx^j\right) = t^i \alpha_j \, T\left(\frac{\partial}{\partial x^i}, dx^j\right)$$

$$= t^i \alpha_j T_i^j$$

$$\Rightarrow \quad T'^k_l = T\left(\frac{\partial}{\partial x'^l}, dx'^k\right) = \frac{\partial x'^k}{\partial x^i}\frac{\partial x^j}{\partial x'^l} T^i_j$$

Př

$$\vec{t} \cdot dx^i = dx^i \cdot \vec{t} = \vec{t}(x^i) = t^i$$

Metrika

$$g = g_{ij} dx^i dx^j$$

- tenzorové pole typu $(0,2)$ – 2-krát kovariantní! tenzor je v každém bodě nedegenerovaný, pozitivně definitní, symetrický

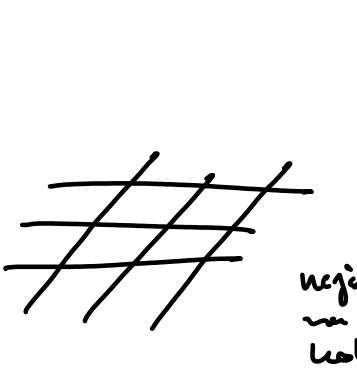
1) $\mathbb{E}^3$ + kartézské souřadnice $\quad g_{ij} = \delta_{ij}$

$\Rightarrow$ vektory ortonormální $\quad \delta'_{ij} = \frac{\partial x^k}{\partial x'^i}\frac{\partial x^l}{\partial x'^j}\delta_{kl}$

$\Rightarrow \quad A^T A = \mathbb{1}$

2) $\mathbb{E}^3$ + lin. souřadnice

g_{ij} = konst



$\Rightarrow$ nemusí být diagonální!

3) $\mathbb{E}^3$ + ortogonální souřadnice

$$g_{ij} = \begin{pmatrix} h_1^2 & & 0 \\ & h_2^2 & \\ 0 & & h_3^2 \end{pmatrix} \qquad h_i \dots \text{Lamého koeficienty}$$

$$h_i^2 = g\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^i}\right)$$

4) ortonormální souřadnice – trička

ortogonální $\to$ $\vec{e}_i = \frac{1}{h_i}\frac{\partial}{\partial x^i}$ (nesčítá se !)

$$g(\vec{e}_i, \vec{e}_j) = \delta_{ij}$$

Konstrukce souřadnic

Dim = 2

$(x,y) \to z = x + iy \to f(z) = u(x,y) + i v(x,y)$

$\mathbb{E}^2$

Předpoklady:

1) $f'(z) = \frac{df}{dz}(z) \neq 0$

2) f je holomorfní $\to$ konformní zobrazení

Cauchy Riemann $\quad u_{,x} = v_{,y}$, $\quad u_{,y} = -v_{,x}$

$\downarrow$ zachovává úhly

$df \leftrightarrow \nabla f$

$$g^{-1}(du, dv) = g^{-1}(u_{,x}dx + u_{,y}dy, \; v_{,x}dx + v_{,y}dy)$$

kartézská metrika = inverzní

$$= u_{,x}v_{,x} + u_{,y}v_{,y}$$

$$= -u_{,y}u_{,x} + u_{,y}u_{,x} = 0$$

opačný přístup – euklidovská geometrie funguje v u a v

$$\frac{\partial}{\partial x} = \frac{\partial u}{\partial x}\frac{\partial}{\partial u} + \frac{\partial v}{\partial x}\frac{\partial}{\partial v}$$

$$\frac{\partial}{\partial y} = \frac{\partial u}{\partial y}\frac{\partial}{\partial u} + \frac{\partial v}{\partial y}\frac{\partial}{\partial v}$$

$$g\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right) = \frac{\partial u}{\partial x}\cdot\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\cdot\frac{\partial v}{\partial y} = 0$$

Př

$z \to e^z$

$$x + iy \to e^x e^{iy} = e^x(\cos y + i \sin y)$$

$$u = e^x \cos y \qquad v = e^x \sin y$$

$\Rightarrow$ odpovídá transformaci do tak trochu polárních souřadnic

z x = konst. jsou kružnice

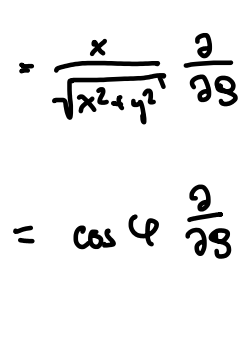
z y = konst. jsou polopřímky z počátku

Kartézské $\to$ polární

$\mathbb{E}^3$ 2d souřadnice $\to$ stočení (polární $\to$ cylindrické)

$\to$ rotace (polární $\to$ sférické)

2d eliptické souřadnice



$\to$ protáhlé elipsoidální souřadnice

$\to$ zploštělé elipsoidální souřadnice

Obecněji v $\mathbb{E}^3$

- trojosý elipsoidální souřadný systém

Cylindrické souřadnice

$x = \rho\cos\varphi \qquad \rho = \sqrt{x^2+y^2}$

$y = \rho\sin\varphi \qquad \varphi = \operatorname{arctg}\frac{y}{x}$

$z = z$

$dx = \cos\varphi \, d\rho - \rho\sin\varphi \, d\varphi$

$dy = \dots$

$dz = dz$

$$\frac{\partial}{\partial x} = \frac{\partial \rho}{\partial x}\frac{\partial}{\partial \rho} + \frac{\partial \varphi}{\partial x}\frac{\partial}{\partial \varphi}$$

$$= \frac{x}{\sqrt{x^2+y^2}}\frac{\partial}{\partial \rho} - \frac{y}{x^2+y^2}\frac{\partial}{\partial \varphi} \Big\} \text{ v hybridní fázi}$$

$$= \cos\varphi \frac{\partial}{\partial \rho} - \frac{\sin\varphi}{\rho}\frac{\partial}{\partial \varphi}$$

$$g = dx\,dx + dy\,dy + dz\,dz$$

$$= (\cos\varphi\, d\rho - \rho\sin\varphi\, d\varphi)(\cos\varphi\, d\rho - \rho\sin\varphi\, d\varphi)$$

$$+ (\sin\varphi\, d\rho + \rho\cos\varphi\, d\varphi)(\sin\varphi\, d\rho + \rho\cos\varphi\, d\varphi)$$

$$+ dz\,dz$$

$$= d\rho\,d\rho + \rho^2 d\varphi\,d\varphi + dz\,dz$$

$$-\rho\sin\varphi\cos\varphi\,(\cancel{d\varphi\,d\rho} + \cancel{d\rho\,d\varphi} - \cancel{d\varphi\,d\rho} - \cancel{d\rho\,d\varphi})$$

$$= \underline{1}\, d\rho\,d\rho + \underline{\rho^2}\, d\varphi\,d\varphi + \underline{1}\, dz\,dz$$

Lamého koeficienty

$h_\rho = 1 \qquad h_\varphi = \rho \qquad h_z = 1$

Sférické souřadnice

- analogicky:

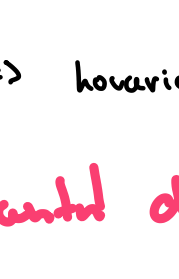
$${}^{3D}g = dr\,dr + r^2\,d\theta\,d\theta + r^2\sin^2\theta\,d\varphi\,d\varphi$$

$r = 2 \to$ 2D = sféra

$\Rightarrow dr = 0$

$\Rightarrow {}^{2D}g = d\theta\,d\theta + \sin^2\theta\,d\varphi\,d\varphi$

$\to$ porovnej s polárními souřadnicemi



$\to$ po přechodu měří $\to$ jinam

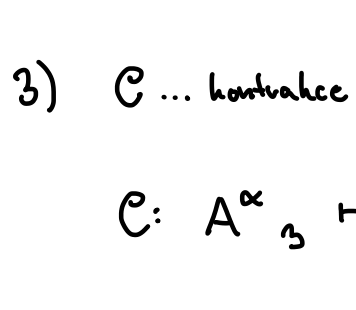
Derivování tenzorů

$$T = T^{ab}_{cd}\frac{\partial}{\partial x^a}\frac{\partial}{\partial x^b}dx^c dx^d$$

obecně nejsou homogenní pole

$\Rightarrow$ při derivování se tyto musí zderivovat

vezmi:



$$\frac{dT(\gamma(s))}{ds}\Big|_{s=0} = \vec{t}\cdot\nabla T$$

$$\lim_{\delta\to 0}\frac{T(\gamma(s-\delta)) - T(\gamma(s))}{\delta}$$

} ale odečítat tenzory ve dvou bodech je nemožné!

$\Rightarrow$ vede na zeď!

$\to$ musíme na to jinak

derivací tenzoru by měl být opět tenzor

$\Rightarrow$ kovariantní derivace axiomaticky

Kovariantní derivace axiomaticky

1) $\nabla(A+B) = \nabla A + \nabla B$

A, B jsou tenzory stejného typu

2) $\nabla(AB) = (\nabla A)B + A(\nabla B)$

z hlediska pozice ideální tenzorový

speciálně

$$\nabla(fA) = (\nabla f)A + f(\nabla A)$$

3) $\mathcal{C}$ … kontrakce

$$\mathcal{C} : A^\alpha{}_\beta \mapsto A^\alpha{}_\alpha$$

$$\nabla(\mathcal{C}A) = \mathcal{C}\,\nabla A$$

4) ∇f … gradient

Takovýchto operací je nekonečně mnoho

Př

$\mathbb{E}^d$ … lineární souřadnice

$$\nabla dx^{\bar{m}} = 0 \iff \nabla\frac{\partial}{\partial x^{\bar{m}}} = 0$$

lineární souřadnice

$$\nabla T = \nabla\left(T^{\bar{a}\bar{b}\dots}_{\bar{c}\bar{d}\dots}\frac{\partial}{\partial x^{\bar{a}}}\frac{\partial}{\partial x^{\bar{b}}}\dots dx^{\bar{c}}dx^{\bar{d}}\dots\right)$$

$$= \left(\nabla T^{\bar{a}\bar{b}}_{\bar{c}\bar{d}}\right)\frac{\partial}{\partial x^{\bar{a}}}\frac{\partial}{\partial x^{\bar{b}}}\dots dx^{\bar{c}}dx^{\bar{d}}\dots)$$

$$= T^{\bar{a}\bar{b}}_{\bar{c}\bar{d}\dots,\bar{k}}\,dx^{\bar{k}}\frac{\partial}{\partial x^{\bar{a}}}\frac{\partial}{\partial x^{\bar{b}}}\dots dx^{\bar{c}}dx^{\bar{d}}\dots$$

Pozn.

složky tenzoru

$$a^i_j\,dx^j \otimes \frac{\partial}{\partial x^i}$$

$$dx^i \cdot \frac{\partial}{\partial x^j} = \frac{\partial x^i}{\partial x^j} = \delta^i_j$$

$$\vec{a}\cdot\underset{\sim}{\alpha} = \left(a^j\frac{\partial}{\partial x^j}\right)\cdot(\alpha_i dx^i) = a^j\alpha_i\frac{\partial}{\partial x^j}\cdot dx^i = a^j\alpha_j$$

Kovariantní derivace

$$\nabla\vec{a} = a^i{}_{;j}\,dx^j\frac{\partial}{\partial x^i} = (\nabla\vec{a})^i{}_j\,dx^j\frac{\partial}{\partial x^i}$$

$$= \nabla_j a^i\,dx^j\frac{\partial}{\partial x^i}$$

} značení

Parametrizace

$$\nabla\vec{a} = \nabla\left(a^i\frac{\partial}{\partial x^i}\right) = (\nabla a^i)\frac{\partial}{\partial x^i} + a^i\left(\nabla\frac{\partial}{\partial x^i}\right)$$

tady musí být typu (1,1) aby tvořil tenzorový zápis

$$= a^i{}_{,j}\,dx^j\frac{\partial}{\partial x^i} + a^i\Gamma^k_{ji}\,dx^j\frac{\partial}{\partial x^k}$$

$$= \underbrace{\left(a^i{}_{,j} + a^l\Gamma^i_{jl}\right)}_{=:\; a^i{}_{;j}}dx^j\frac{\partial}{\partial x^i}$$

$$\nabla\underset{\sim}{\alpha} = \nabla(\alpha_j dx^j) = \alpha_{j,i}\,dx^i dx^j - \alpha_k\tilde{\Gamma}^k_{ij}\,dx^i dx^j$$

záleží na pořadí!

$$= \underbrace{\left(\alpha_{j,i} - \alpha_k\tilde{\Gamma}^k_{ij}\right)}_{=:\;\alpha_{j;i}}dx^i dx^j$$

$$0 = \nabla\left(dx^i\cdot\frac{\partial}{\partial x^j}\right) = -\tilde{\Gamma}^i_{kl}\,dx^k \overbrace{dx^l\cdot\frac{\partial}{\partial x^j}}^{\delta^l_j}$$

$$+ \underbrace{dx^i\cdot\Gamma^l_{kj}\,dx^k\frac{\partial}{\partial x^l}}_{\delta^i_l}$$

$$= -\tilde{\Gamma}^i_{kj}\,dx^k + \Gamma^i_{kj}\,dx^k$$

$$\Rightarrow \quad \tilde{\Gamma} = \Gamma \qquad \text{Christoffelovy symboly}$$

$$T^{ab}_{cd;j} = T^{ab}_{cd,j} + \Gamma^a_{jk}T^{kb}_{cd} + \Gamma^b_{jk}T^{ak}_{cd}$$

$$- \Gamma^k_{jc}T^{ab}_{kd} - \Gamma^k_{jd}T^{ab}_{ck}$$

$$\frac{\partial}{\partial x^j}\cdot\left(\nabla\frac{\partial}{\partial x^i}\right)\cdot dx^k = \Gamma^k_{ji}$$

$$\frac{\partial}{\partial x^i}\cdot(\nabla dx^j)\cdot\frac{\partial}{\partial x^k} = \Gamma^j_{ki}$$

Př $\quad \mathbb{E}^d$ a lineární souřadnice $x^{\bar{i}}$

$$\nabla dx^{\bar{i}} = 0 = 0\,\nabla\frac{\partial}{\partial x^{\bar{i}}}$$

$$\Gamma^k_{ji} = \frac{\partial}{\partial x^j}\cdot\left(\nabla\frac{\partial x^{\bar{m}}}{\partial x^i}\right)\frac{\partial}{\partial x^{\bar{m}}}\cdot dx^k$$

$$= \frac{\partial}{\partial x^j}\cdot\frac{\partial^2 x^{\bar{m}}}{\partial x^n\partial x^i}\frac{\partial}{\partial x^m}dx^n\frac{\partial}{\partial x^{\bar{m}}}\cdot dx^k$$

$$= \frac{\partial^2 x^{\bar{m}}}{\partial x^j\partial x^i}\frac{\partial x^k}{\partial x^{\bar{m}}}$$

$$\Gamma^j_{ki} = \frac{\partial}{\partial x^i}(\nabla dx^j)\frac{\partial}{\partial x^i} = \frac{\partial}{\partial x^k}\left(\nabla\frac{\partial x^j}{\partial x^{\bar{m}}}dx^{\bar{m}}\right)\frac{\partial}{\partial x^i}$$

$$= \frac{\partial^2 x^j}{\partial x^k\partial x^{\bar{m}}}\frac{\partial x^m}{\partial x^j} = \frac{\partial^2 x^i}{\partial x^{\bar{l}}\partial x^{\bar{m}}}\frac{\partial x^{\bar{l}}}{\partial x^k}\frac{\partial x^{\bar{m}}}{\partial x^i}$$

Obecný případ

1) Symetrie $\quad \Gamma^i_{jk} = \Gamma^i_{kj}$ ≡ beztorznost

2) kompatibilita s metrikou $\quad \nabla g = 0$

$$((\nabla g)_{abc}\,dx^c dx^a dx^b)$$

$$0 = g_{ab;c} = g_{ab,c} - \Gamma^d_{ca}g_{db} - \cancel{\Gamma^d_{cb}g_{ad}} \quad ⊖$$

$$0 = g_{bc;a} = g_{bc,a} - \Gamma^d_{ab}g_{dc} - \cancel{\Gamma^d_{ac}g_{bd}} \quad ⊕$$

$$0 = g_{ca;b} = g_{ca,b} - \cancel{\Gamma^d_{bc}g_{da}} - \Gamma^d_{ba}g_{cd} \quad ⊕$$

$$\Gamma^d_{ab}g_{dc} = g_{bc,a} + g_{ca,b} - g_{ab,c}$$

$$\Gamma^f_{ab} = \frac{1}{2}g^{cf}(g_{bc,a} + g_{ca,b} - g_{ab,c})$$

Speciálně v $\mathbb{E}^d$ s ortogonálními souř.

$$g = \begin{pmatrix} h_1^2 & & 0 \\ & h_2^2 & \\ 0 & & \ddots \; h_d^2 \end{pmatrix}$$

$k \neq m \neq n \neq k$

$$\Gamma^k_{mn} = \frac{1}{2}g^{kk}(0 + 0 - 0) = 0$$

$k \neq m$

$$\Gamma^k_{mm} = \frac{1}{2}g^{kk}(0 + 0 - g_{mm,k})$$

$$= -\frac{1}{2}\frac{1}{h_k^2}(h_m^2)_{,k} = -\frac{h_m h_{m,k}}{h_k^2}$$

$$\Gamma^k_{km} = \frac{1}{2}g^{kk}(g_{km,k} + g_{kk,m} - g_{km,k})$$

$$= \frac{h_k h_{k,m}}{h_k^2}$$

$$\sum_k\Gamma^k_{km} = \Big[\ln\underbrace{(h_1\cdot h_2\cdot h_3\cdots)}_{\sqrt{\det g}}\Big]_{,m}$$

Pozn. ON báze na $\mathbb{E}^d$

$$e_{\hat{i}} = \frac{1}{h_i}\frac{\partial}{\partial x^i} \quad (\text{nesčítá se})$$

$$e^{\hat{j}} = h_j\,dx^j$$

$$\vec{a} = a^i\frac{\partial}{\partial x^i} = a^{\hat{i}}e_{\hat{i}}$$

$$h_i a^i = a^{\hat{i}} = a_{\hat{i}} = \frac{1}{h_i}a_i$$

Operátory v ortog(on)onální souř.

$$(\nabla f)_m = \nabla_m f = f_{,m} \Rightarrow (\nabla f)_{\hat{m}} = \frac{f_{,m}}{h_m} = (\nabla f)^{\hat{m}}$$

$$\nabla\cdot\vec{a} = \nabla_i a^i = a^i{}_{;i} = a^i{}_{,i} + \underbrace{\Gamma^i_{ij}}_{\text{sčítá se}}a^j$$

$$= a^j{}_{,j} + \frac{(h_1\cdots h_d)_{,j}}{h_1\cdots h_d}a^j$$

$$= \frac{(a^j h_1\cdots h_d)_{,j}}{h_1\cdots h_d}$$

speciálně $d = 3$

$$\Rightarrow \quad \nabla\cdot\vec{a} = \frac{1}{h_1h_2h_3}\left[(a^1h_2h_3)_{,1} + (a^2h_1h_3)_{,2} + (a^3h_1h_2)_{,3}\right]$$

kovariantní

$$= \frac{1}{h_1h_2h_3}\left[\left(\frac{a_1}{h_1}h_2h_3\right) + \dots\right]$$

$$\Delta f = \nabla\cdot\nabla f = f^{;k}{}_{;k} = f_{;k}{}^{;k} =$$

můžu manipulovat s metrikou, jelikož $\nabla g = 0$

$$= \frac{1}{h_1h_2h_3}\left[\left(\frac{h_2h_3}{h_1}f_{,1}\right)_{,1} + \left(\frac{h_1h_3}{h_2}f_{,2}\right) + \left(\frac{h_1h_2}{h_3}f_{,3}\right)_{,3}\right]$$

Vsuvka

Levi-Civitův symbol

$$\varepsilon \in \mathcal{T}^0_d(\mathbb{E}^d)$$

1) antisymetrický kompletně

$$\varepsilon_{i_1\dots i_d} = \varepsilon_{[i_1\dots i_d]}$$

má plný rank

2) normalizace

$$\varepsilon_{\hat{1}\dots\hat{d}} = 1 \quad \text{pro pravotočivě orientované báze}$$

$$\varepsilon_{i_1\dots i_d}\,\varepsilon^{i_1\dots i_d} = d!$$

$\Rightarrow$ co platí pro $\varepsilon_{1\dots d}$?

$$\varepsilon_{1\dots d} = h_1\cdot\dots\cdot h_d$$

$$\varepsilon^{1\dots d} = \frac{1}{h_1}\cdot\dots\cdot\frac{1}{h_d}$$

$\varepsilon_{1\dots d}\,dx^1\dots dx^d$ $\to$ zajišťuje orientaci integračního elementu

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$$(\nabla\times a)^k = \varepsilon^{kmn}a_{n;m} = \varepsilon^{kmn}\big(a_{n,m} - \underbrace{\Gamma^j_{nm}a_j}_{\to 0}\big)$$

antisym. · sym.

$$= \varepsilon^{kmn}a_{n,m}$$

$$(\nabla\times a)^1 = \varepsilon^{123}(a_{3,2} - a_{2,3}) = \frac{1}{h_1h_2h_3}(a_{3,2} - a_{2,3})$$

$$= \frac{1}{h_1h_2h_3}\left((h_3^2a_3)_{,2} - (h_2^2a_2)_{,3}\right)$$

$$(\nabla\times a)^{\hat{1}} = \frac{1}{h_2h_3}\left((h_3a_{\hat{3}})_{,2} - (h_2a_{\hat{2}})_{,3}\right)$$